Fiscal Year:	FY 2007	**Task Last Updated:**	FY 02/16/2008
PI Name:	Beckingham, Kathleen M. Ph.D.		
Project Title:	Drosophila Behavior and Gene Expression in Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	77005	**Congressional District:**	25
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Space Biology ILSRA
Start Date:	08/01/2002	**End Date:**	06/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	12	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	NOTE: received NCE through 6/30/2007, per A. Chu (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bhattacharya, Sharmila (NASA Ames Research Center)		
Grant/Contract No.:	NCC2-1356		
Performance Goal No.:			
Performance Goal Text:			
	Model organisms are of great value in deciphering the fundamental mechanisms underlying universal biological processes. The unparalleled genetic methodologies available for the fruit fly Drosophila melanogaster make it uniquely advantageous as a model system for the study of gravity perception. Drosophila has been adopted as the insect of choice for studies in the Insect Habitat of the International Space Station. The experiments proposed derive from the long-term objective of using Drosophila to understand the molecular processes of gravity perception. It is proposed to examine the effects of microgravity, and other gravitational force levels, on behavior and gene expression in Drosophila. Preliminary studies have identified a period of adaptation to hypergravitational force that is characterized by aberrant behavior, followed by resumption of normal activity. We will determine whether similar changes occur in microgravity and determine whether gene expression is altered during the critical adaptation period. The complete sequencing of the		

Task Description:	Drosophila genome and generation of microarrays representing essentially all genes within the organism will allow us to examine gene expression on a global level and identify all genes affected by gravitational force change. Preliminary work has also identified mutant Drosophila with abnormal gravitational sensing. Mutants for which the affected gene as been identified and proved i) to be novel ii) to be expressed in sense organs indicated to detect gravity and iii) to have a human homolog, will also be used in these studies. Determination of the gravity-induced behavioral and gene expression changes in these mutant files and comparison to the wild type parameters will address the fundamental roles of the affected genes in gravity perception and the relevance of any gene expression changes seen in wild type files.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A deeper understanding of the molecular mechanisms underlying graviperception.
Task Progress:	Prior to the initiation of this funding we had undertaken a large scale genetic screen to identify Drosophila mutants with abnormal responses to gravity. The screening device used was a so-called "gravitaxic maze" - that is, a vertical maze with nine choice points at which flies can choose to walk up or down. Flies thus leave the maze at one of nine vertically positioned exits depending on their aggregate choices. In using this screening device we selected mutants that left the maze at unusually high or unusually low exit points when compared to appropriate controls. Further behavioral testing was performed to investigate the specificity of the behavioral defect in each mutant line. The goal of the studies undertaken with this funding was to characterize these mutants further so that a small number of ideal candidates could be selected for study in microgravity. The plan was to study the behavior of these mutants as compared to controls once exposed to microgravity and also to compare the changes in their patterns of gene expression in microgravity to those seen for control strains. Further, experiments with various habitat chambers were planned with a view to generating ideal growth and maintenance conditions for use in microgravity. Towards these goals, 23 mutants were identified as showing very specific changes in gravity-related behavior while other behaviors such as flight and mating behavior were normal. The genes affected by these mutations were then cloned. Interestingly, many of these genes proved to encode proteins predicted to have roles in neural signaling or development. Further, several of these genes proved to be highly conserved in mammals, suggesting that the molecular pathways involved in gravity sensing are maintained across evolution. Six genes were selected for detailed analysis. Five of these genes are novel Drosophila genes and thus we were given the task of naming these genes. We chose the names of some of the best known space explorers of our age - yuri gagarin, alan shepard, john glenn, jim lovell, and neil armstrong. The sixth gene represents the Drosophila version of a gene previously identified in mammals and recognized as being expressed in brain regions involved in locomotor control. However, no mutants for this gene, which is called Striatin, have been identified in mammalian species and thus our mutant provides the first opportunity for genetic analysis of the function of this gene. Our further work with these genes has involved characterization of their functions in the organism through generation of additional mutations and detailed molecular studies. The gene yuri gagarin appears to control gravity behavior through its role in the putative gravity sensor organ of Drosophila, termed Johnston's Organ. This structure is present in the second antennal segment. Sensation of gravity in this structure al most certainly depends upon deformation of the cilia of certain mechano-sensory neurons. The gene product of the yuri gene appears to essential for the proper functioning of these cilia. The genes alan shepard and jim lovell both appear to have roles in the development of the nervous system with roles in axon pathfinding. Both genes are expressed in neurons that form the equivalent in Drospohila of the vertebrate spinal cord. Additional mutants in these genes are proving useful in determining the exact roles of these genes in these vital processes. It seems likely that the gravitaxic defects detected with the initial mutations reflect alterations in the development of the neural circuitry required to execute coordinated gravitational responses. Our studies of the three further genes selected for study are less well advanced but in these cases also, early studies indicate roles in the signaling, or the development of, essential neural circuitry. Parallel studies in collaboration with Dr. Sharmila Bhattacharya at NASA Ames focused on determining the best conditions for rearing both the mutant and control flies for prolonged periods in microgravity. Containers suitable for the EMCS hardware were ultimately prepared. The effects of hypergravity of some of our mutants were also investigated in collaboration with Dr. Bhattacharya. In collaboration with Dr. J. Doug Armstrong, now at the University of Edinburgh, development of software suitable for monitoring the behavior of the flies on transition to microgravity was initiated. We are attempting to continue with our analysis of these various Drosophila mutants despite the cessation of the funding for this project. The rich genetic analysis possible in Drosophila offers the potential for understanding elements of gravity perception that are not possible in any other animal species.
Bibliography Type:	Description: (Last Updated: 03/14/2017)
Articles in Peer-reviewed Journals	Armstrong JD, Texada MJ, Munjaal R, Baker DA, Beckingham KM. "Gravitaxis in Drosophila melanogaster: a forward genetic screen." Genes Brain Behav. 2006 Apr;5(3):222-39. PMID: 16594976 , Apr-2006
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Journal/Magazine covers	Texada MJ, Simonette RA, Johnson CB, Deery WJ, Beckingham KM. "Cover in Journal of Cell Science for article, 'yuri gagarin is required for actin, tubulin and basal body functions in Drosophila spermatogenesis.' " J Cell Sci. 2008 Jun 1;121(Pt 11):1926-36. PMID: 18477609 , Jun-2008